

Environmental Impacts of Deforestation

A structured overview of the environmental, economic and social consequences of deforestation

Environmental Impacts

A. Loss of Biodiversity

Point:

Deforestation **destroys the habitats** of plants and animals, resulting in loss of biodiversity.

Explanation:

This refers to the **loss of the variety** of plants and animals. With the **loss of biodiversity**, the **variety of resources** that can be **obtained from the rainforest**, such as **food, medicine and timber** wood, will also be **gone**.

Example:

Researchers in 1991 isolated a compound which **blocked the spread of the AIDS virus in human cells**. The team **sent biologists to Malaysia for more samples from the Malaysian Gum Tree**, but the **tree had been cut down** and **no other tree had been found** that **produced the same compound**.

Link:

Thus, **deforestation destroys the habitats of plants and animals**, resulting in loss of biodiversity.

B. Loss of Water Catchment Areas

Point:

Forests play an important role in **maintaining our water supply**; they **enable water** to be **collected and stored within a water catchment area**.

Explanation:

Clearing trees **disrupts the water cycle**. Trees **absorb water through their roots and release it into the atmosphere** through **evaporation and transpiration**. The **water in the atmosphere** eventually **falls again as rain**. When trees are removed, the **water cycle is affected**, leading to a **drier climate and drier soil**. Hence, the **ground no longer contains and receives as much water**.

Example:

In **Tanzania**, **deforestation caused by charcoal production is predicted to reduce half of the water catchment area by 2020**. The **cost of water is expected to rise by four times**.

Link:

Thus, **deforestation causes the loss of water catchment areas**.

C. Increased Risk of Flooding, Soil Erosion and Sedimentation

Point:

Tree roots hold the soil together and break the fall of rainwater.

Explanation:

When deforestation occurs, there are no longer any tree roots to hold the soil together. This means that the soil can be easily blown away or washed off by wind or rainwater. When the

eroded soil particles enter a nearby river, it increases the amount of sediment in the water. This can smother fish eggs and coral, and lower water quality.

Example:

The Great Red Island of Madagascar has lost so much soil to erosion that its rivers run blood-red, affecting even the Indian Ocean.

Explanation:

When trees are cleared, rainwater falls directly on the ground and flows over the surface more rapidly as surface runoff, causing flash floods in the surrounding areas.

Example:

Rampant logging along the Batang Ayum River in Sumatra, Indonesia, has resulted in deadly flash floods in recent years.

Link:

Thus, deforestation increases the risk of flooding, soil erosion and sedimentation.

D. Enhanced Greenhouse Effect

Point:

Trees store carbon and photosynthesise, which absorbs carbon dioxide and releases oxygen.

Explanation:

Deforestation increases the amount of carbon dioxide, a greenhouse gas, in the atmosphere. This is because trees absorb carbon dioxide from the atmosphere and store it as carbon, and when they are cut down, they release it into the environment.

Example:

More than half of the carbon dioxide released due to deforestation comes from Brazil and Indonesia, where large areas of tropical rainforest are found.

Link:

Thus, deforestation enhances the greenhouse effect as carbon is released.

Economic Impact

Point:

Tropical rainforests provide natural resources that contribute greatly to the economic growth of many countries.

Explanation:

With rapid deforestation, the rainforest is destroyed faster than it can replenish itself and hence will not be able to provide these resources for many industries that need them.

Example:

It is estimated that the amount of tropical rainforest destroyed around the world each year amounts to US\$45 billion.

Link:

Thus, deforestation depletes natural resources.

Social Impact

Point:

Deforestation causes indigenous people to lose their homes and cultures.

Explanation:

The displacement from their homes often causes them to disperse, and their culture and way of life will be lost.

Example:

The Guarani community in Brazil had to live in makeshift camps in Mato Grosso when their homes in the Amazon rainforest were destroyed following rapid deforestation to build sugarcane plantations.

Link:

Thus, deforestation affects indigenous people greatly.